Reference states for available potential energy in AN2D and FC2D: algorithms

Notes on the `TraceableEnergetics` code

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Scope. This note specifies how to compute the *reference state* of a snapshot of AN2D or FC2D, precisely enough to be implemented. The available energy density, its budget and the conditions that the reference state must satisfy are derived in the energy-budget notes `energy.tex` ("Energy budgets for compressible and anelastic fluids", "the energy notes" below). Section 1 summarises what they require, and §11 lists what this note delivers to them. Notation and code names follow `traceable_energetics.tex` ("the main note"): ϑ is the entropic variable (`consvar`), q the solute fraction, $h(P,\vartheta,q)$ the specific enthalpy with $\partial h/\partial P|_{\vartheta,q} = v$, $\Phi = g(z - z_0)$, and the box is $[0, L_x] \times [0, L_z]$ with $M_x \times M_z$ interior cells of size $\Delta x \times \Delta z$.

Correspondence with the energy notes. The energy notes write the pressure p and $h(\vartheta, q, p)$. Here, as in the main note, P is the pressure and $p = (P - P_R)/\rho_R$ the reduced pressure of AN2D. Their static energy is

$$\Sigma(\vartheta, q, z, P) = \begin{cases} h(P, \vartheta, q) + \Phi(z) & \text{FC2D}, \\ h(P_R(z), \vartheta, q) + \dfrac{P - P_R(z)}{\rho_R(z)} + \Phi(z) & \text{AN2D} \end{cases} \tag{1}$$

(they take $z_0 = 0$; the available energy does not depend on z_0). Reference-state quantities carry a subscript $*$, as there: the reference pressure $P_*(z,t)$, the reference height $z_*(\vartheta,q,t)$ of a parcel, the reference specific volume $v_*(z,t)$, the reference density $\rho_* = 1/v_*$ (FC2D) and $N_*^2(\vartheta,q,t)$. In AN2D the reduced reference pressure is $\pi_* = (P_* - P_R)/\rho_R$.

Two families of reference state are covered:

1. the **Lorenz reference state**: the state of minimum potential energy reachable by an adiabatic rearrangement of the parcels (§3–7). Four solvers are specified: sorting, greedy filling with repair, exact rearrangement, and histogram (PDF) reduction;
2. the **horizontal-mean reference state**: a profile that is not a rearrangement of the fluid (§9).

In both, parcels are given their reference height by the level-of-neutral-buoyancy (LNB) equation against the reference profile (§8). Section 13 records the design decisions, including an accepted discretisation mismatch in FC2D, and §14 lists the tests.

1 What the reference state must satisfy

The energy notes define the available energy (AE) of a parcel at height z and pressure P,

$$\Pi = \Sigma(\vartheta, q, z, P) - \Sigma\left(\vartheta, q, z_*, P_*(z_*, t)\right) - \left(P - P_*(z,t)\right) \Sigma_P(\vartheta, q, z, P), \tag{2}$$

with $\Sigma_P = v$ (FC2D) or $1/\rho_R$ (AN2D). It splits as $\Pi = \Pi_1 + \Pi_2$, the available compressible energy (ACE) $\Pi_1 = \Sigma(\vartheta,q,z,P) - \Sigma(\vartheta,q,z,P_*(z,t)) - (P - P_*(z,t))\,\Sigma_P(\vartheta,q,z,P)$, zero in AN2D, and the available potential energy (APE) $\Pi_2 = \Sigma(\vartheta,q,z,P_*(z,t)) - \Sigma(\vartheta,q,z_*,P_*(z_*,t))$. APE always means Π_2 here; the energy released by the Lorenz rearrangement of the whole box is written $E_P - E_*$. The reference profile defines v_* by

$$v_* \frac{\partial P_*}{\partial z} = -g \quad \text{(FC2D)}, \qquad \frac{\partial \pi_*}{\partial z} = g\left(\rho_R v_* - 1\right) \quad \text{(AN2D)}. \tag{3}$$

The conditions on P_* and z_* (energy notes, "Reference state" and the appendix "Variational origin of the reference state conditions") are:

1. **Level of neutral buoyancy (LNB).** $\partial_z \Sigma(\vartheta, q, z, P_*(z,t)) = 0$ at $z = z_*$, i.e. $v(P_*(z_*,t), \vartheta, q) = v_*(z_*,t)$ (FC2D) or $v(P_R(z_*), \vartheta, q) = v_*(z_*,t)$ (AN2D).

2. **Maximal APE.** $N_*^2 = \partial_z^2 \Sigma(\vartheta, q, z, P_*(z,t)) > 0$ at $z = z_*$.

3. **Volume constraint.** For every $f(z)$,

$$\int f(z_*)\, \Sigma_P\left(\vartheta, q, z_*, P_*(z_*,t)\right) \mathrm{d}m = \int f(z)\, \mathrm{d}V, \tag{4}$$

   i.e. the reference state fills the box level by level. In AN2D $\Sigma_P = 1/\rho_R(z_*)$, so the reference state has the mass per unit height $L_x \rho_R(z)$ of the model.

These are the stationarity conditions of $\int \Pi \,\mathrm{d}m$ with respect to the reference state. At fixed P_*, Π depends on z_* only through $-\Sigma(\vartheta, q, z_*, P_*(z_*,t))$, so (1) and (2) are the first- and second-order conditions for z_* to maximise the parcel's Π. Condition (3) is stationarity with respect to P_*, which is the Lagrange multiplier of the volume constraint (exactly so in AN2D; in FC2D $\int \Pi \,\mathrm{d}m$ is a minimum with respect to P_*). When (3) holds, $\int \Pi \,\mathrm{d}m = E_P - E_*$, the potential energy of the actual state minus that of the rearranged state (Lorenz). With the choice of z_* below, the rearranged state is then the one of minimum energy (§3.5).

Choice of z_*. This note takes z_* to be the *global* minimiser of $z \mapsto \Sigma(\vartheta, q, z, P_*(z,t))$ on $[0, L_z]$, the height that maximises the parcel's Π. It is a function of (ϑ, q, t) only, as in the energy notes, and it satisfies (1)–(2) whenever it is interior. Section 8.2 gives the algorithm and compares it with the rule of Tailleux (2013).

What the budgets need. Beyond the model state, the AE budgets of the energy notes need z_* for each parcel, $P_*(z,t)$ and $v_*(z,t)$ at the model rows and at z_*, and $\partial_t P_*(z,t)$ (§10). The terms in $\partial_t z_*$ drop out of the budgets by the LNB condition, so z_* need not be followed in time. This holds only if z_* minimises $\Sigma(\vartheta, q, z, P_*(z,t))$ for the same profile P_* that is used to evaluate Π (§8.1).

The discrete problem. Section 3 poses the Lorenz state as an optimal rearrangement of the parcels between levels. Its placement is the discrete z_*, its slot constraints are the discrete volume constraint, and its dual potentials are the discrete P_* (§3.5). The rearrangement problem and its dual are the discrete form of the variational principle above.

2 Why the problem is not just a sort

For a single-component fluid, or a Boussinesq fluid with an equation of state linear in its tracers, the Lorenz state is obtained by sorting the parcels by a single conserved quantity (Winters et al. 1995). The fluid here is binary and its equation of state (main note, Appendix A) is

$$v = v_0 \left[1 + \alpha_T \left(1 + \gamma(P - P_0)\right)(\vartheta - T_0) - \alpha_q (q - q_0) - \alpha_P (P - P_0) + \tfrac{1}{2}\alpha_{TT}(\vartheta - T_0)^2\right]. \tag{5}$$

When sorting is exact. Suppose v is additively separable,

$$v(P, \vartheta, q) = f(\vartheta, q) + w(P). \tag{6}$$

Then $h(P, \vartheta, q) = h(P_0, \vartheta, q) + f(\vartheta, q)\,(P - P_0) + \int_{P_0}^{P} w$, and the cost of placing parcel i at pressure P is $a_i + f_i P +$ (a function of P only). Only the term $\sum f_i P$ depends on the arrangement, and by the rearrangement inequality it is minimised by placing the largest f (lightest parcel) at the lowest pressure (the top). The same holds with unequal masses and split parcels: with the

parcels ordered by f and the levels by pressure, the costs have the Monge property, for which filling the levels in order (the north-west corner rule) solves the rearrangement problem of §3.2 exactly (Hoffman 1963). In FC2D the pressures of the levels also depend on P_{top} (§3.4), but the order does not, so the same placement is optimal for every P_{top}. So **the sort by f is the exact minimiser whenever** (6) **holds**. For (5) this is the case iff $\gamma = 0$: compressibility (α_P), haline contraction (α_q) and cabbeling (α_{TT}) do not break it; **thermobaricity ($\gamma \neq 0$) does**. The experiments use $\gamma = 10^{-7}\,\text{Pa}^{-1}$, so the sort is an approximation. The relative size of the coupling term across the box is $\gamma\,\Delta P \approx \gamma\rho g L_z$ ($\approx 5 \times 10^{-4}$ for $L_z = 0.5\,\text{m}$), so the sort should be very close to exact: only parcels whose specific volumes are nearly tied can be misordered. This must be checked (§4, §6), not assumed.

The general problem. Without (6), the order in which parcels stack depends on the pressure at which they are compared. Finding the minimum is then a linear programme, known in the literature as the *linear assignment* or *transportation* problem (Hieronymus & Nycander 2015), solvable exactly in polynomial time (§6). Heuristics that fill the column level by level can return a "reference" state with *more* energy than the actual state, i.e. a negative $E_P - E_*$ (Harris & Tailleux 2018 found this in 30–40% of atmospheric soundings for top-down filling), so every heuristic must be followed by an energy check (§5).

3 The discrete minimisation problem

3.1 Parcels

Each interior cell (i, j) is a parcel, indexed by a single $n = 1, \ldots, N$, $N = M_x M_z$, with mass and conserved properties

$$m_n = \begin{cases} \rho_{R,j}\,\Delta x \Delta z & \text{AN2D}, \\ m^{\rho}_{ij}\,\Delta x \Delta z & \text{FC2D}, \end{cases} \qquad \vartheta_n = \frac{m^{\vartheta}_{ij}}{\rho}, \quad q_n = \frac{m^{q}_{ij}}{\rho}, \tag{7}$$

with $\rho = \rho_{R,j}$ (AN2D) or m^{ρ}_{ij} (FC2D). The total mass is $M_{\text{tot}} = \sum_n m_n$. A rearrangement keeps each parcel's (m_n, ϑ_n, q_n), which is what adiabatic, isohaline, mass-conserving motion does. The reference state is at rest and fills the box. Its pressure $P_*(z,t)$, hence $v_*(z,t)$, is horizontally uniform, but (ϑ, q) need not be uniform along a level (density compensation; energy notes).

3.2 Slots and the rearrangement problem

In both models the reference state is a stack of horizontal layers spanning L_x. Its natural vertical coordinate is *mass*, because the pressure (FC2D) and the height (AN2D) of a layer are known once the mass on either side of it is known, whatever parcels occupy it. Divide the column into K *slots* $k = 1, \ldots, K$ of mass M_k ($\sum_k M_k = M_{\text{tot}}$), each with a per-unit-mass cost c_{nk} of placing parcel n there. A rearrangement is a placement $x_{nk} \geq 0$, the mass of parcel n placed in slot k:

$$\min_{x \geq 0} \sum_{n,k} x_{nk}\, c_{nk} \qquad \text{subject to} \qquad \sum_k x_{nk} = m_n \;\; \forall n, \qquad \sum_n x_{nk} = M_k \;\; \forall k. \tag{8}$$

This linear programme is known as the transportation problem. When all m_n and M_k are equal and $K = N$ it reduces to the linear assignment problem of Hieronymus & Nycander (2015). With $K \ll N$ shared levels it is the minimum-cost flow problem of Su & Ingersoll (2016). Parcels may be split between slots, which is needed because parcel masses differ. The placement is the discrete z_*: $x_{nk} > 0$ places (part of) parcel n at the level of slot k. The models differ only in their slots and costs.

3.3 AN2D: slots are the model rows

By the volume constraint (4) with $\Sigma_P = 1/\rho_R$, the reference state has the mass per unit height $L_x\rho_R(z)$ of the model. These are also the states that anelastic flow can reach: it satisfies $\nabla\cdot(\rho_R\mathbf{u}) = 0$, so its flow map preserves the measure $\rho_R\,\mathrm{d}A$. So the slots are the model rows, and the thermodynamics is evaluated at $P_R(z)$, as in the model. With one slot per row,

$$K = M_z, \qquad M_k = M_x\,\rho_{R,k}\,\Delta x\Delta z, \qquad c_{nk} = h(P_{R,k}, \vartheta_n, q_n) + g\,(z_k - z_0). \tag{9}$$

The slot resolution is set by the user: with s slots per row ($K = sM_z$), row k is split into s slots of mass M_k/s and height $\Delta z/s$, with the costs at their centres and P_R from the interpolant of §8.1. The statements below on exact discrete consistency hold for $s = 1$. For $s > 1$ the actual state is no longer a placement with exactly the model's energy, and they hold to $O(\Delta z^2)$, as in FC2D. The reference pressure $P_* = P_R + \rho_R\pi_*$ is not P_R: it is the multiplier of the slot constraints (§3.5). The potential energy of a placement, with $\mathcal{U} = \rho_R(h(P_R) + \Phi) - P_R$ of the main note, is

$$E[x] = \sum_{n,k} x_{nk}\,c_{nk} - \sum_k M_x\,P_{R,k}\,\Delta x\Delta z, \tag{10}$$

where the last term does not depend on the placement. The actual state is the feasible placement "each parcel in its own row", and its energy E_P is exactly the discrete $\int\mathcal{U}$ of the model. Hence **the exact solution gives $E_P \geq E_*$ to round-off, with equality when the actual state is itself a minimiser**: a horizontally uniform state that is stable to finite exchanges of parcels (with (6), any horizontally uniform, statically stable state). The reference state is consistent with the model's discrete energy.

Placement height of a parcel. A parcel split over several rows has the mass-weighted height $\hat{z}_n = \sum_k x_{nk}z_k/m_n$. Inside a row the parcels are not ordered by the placement. If a sub-row height is wanted (the analogue of the Winters et al. height), order the parcels within row k by the sort key of §4 and place them by cumulative mass: $\hat{z}_n = z_k - \frac{1}{2}\Delta z + (m_{<n} + \frac{1}{2}x_{nk})/(L_x\rho_{R,k})$, where $m_{<n}$ is the mass in the row below parcel n. The z_* used in Π is the LNB height of §8.2; $\hat{z}$ is used to check it.

Reference profile. The reference state has no single (ϑ, q) per row. Its reduced pressure on the slot centres follows from the slot-mean specific volume

$$v_{*,k} = \frac{1}{M_k}\sum_n x_{nk}\,v(P_{R,k}, \vartheta_n, q_n), \tag{11}$$

hydrostatic with it, $\partial_z\pi_* = -v_*\,\partial_z P_R - g$, with v_* linear in P_R between the slot centres (the trapezoidal rule in the pressure),

$$\pi_{*,k} - \pi_{*,k+1} = -\tfrac{1}{2}\,(v_{*,k} + v_{*,k+1})\,(P_{R,k} - P_{R,k+1}) - g\,(z_k - z_{k+1})\,, \tag{12}$$

exact when v is linear in P, so that the static energy of a neutral fluid is flat at the slot centres. The duals of the slot constraints give $\pi_{*,k} = -\psi_k$ (§3.5), the exact multiplier, for which $\int\Pi\,\mathrm{d}m = E_P - E_*$ exactly when every z_* is at a slot centre ($s = 1$). They are not used for the profile (decision of the user, §13): they change abruptly in time when the parcel split between two slots changes, and a parcel split between two slots has the same static energy at both along the dual profile, so its z_* jumps between them. Both make the budgets of the available energy rough in time and space (the generation G by two orders of magnitude in the thermohaline lock exchange), while the slot volumes change continuously with the state; with the refined z_* the gap $\int\Pi\,\mathrm{d}m - (E_P - E_*)$ is about the same for both profiles. The duals certify the placement (§3.5). Either way π_* is defined up to a constant, which does not affect Π (§11); the code makes its mass-weighted mean over the slots zero.

3.4 FC2D: slots of fixed mass, unknown top pressure

In a compressible reference column the layer thicknesses are not known in advance. What is known is the pressure as a function of mass: if m is the mass above a point (per box, from the top), hydrostatic balance gives $P = P_{\mathrm{top}} + gm/L_x$, whatever the arrangement. Only the top pressure P_{top} is unknown, and it is set by the condition that the parcels fill the box. Take K slots of equal mass $M_k = M_{\mathrm{tot}}/K$, numbered from the top, with

$$P_k(P_{\mathrm{top}}) = P_{\mathrm{top}} + \frac{g}{L_x}\left(k - \tfrac{1}{2}\right)\frac{M_{\mathrm{tot}}}{K}, \qquad c_{nk}(P_{\mathrm{top}}) = h\left(P_k(P_{\mathrm{top}}), \vartheta_n, q_n\right). \tag{13}$$

K is set by the user. $K = M_z$ gives as many layers as model rows; a finer K improves the vertical resolution of the reference profile, at a proportional cost.

Energy of a hydrostatic column. For a column at rest in hydrostatic balance, integrating $\rho g z$ by parts with $\mathrm{d}P = -\rho g\,\mathrm{d}z$ gives

$$\int \rho\,(e + \Phi)\,\mathrm{d}A = \int h\,\mathrm{d}m - P_{\mathrm{top}}\,L_x L_z - g z_0 M_{\mathrm{tot}}. \tag{14}$$

The discrete form uses the midpoint rule within slots, $\sum_{n,k} x_{nk} c_{nk}$.

The saddle problem. For a fixed placement, define $G(x, P_{\mathrm{top}}) = \sum x_{nk} c_{nk}(P_{\mathrm{top}}) - P_{\mathrm{top}} L_x L_z$. Then $\partial G/\partial P_{\mathrm{top}} = \sum x_{nk} v(P_k, \vartheta_n, q_n) - L_x L_z$ is the volume excess, and $\partial^2 G/\partial P_{\mathrm{top}}^2 = \sum x_{nk}\,\partial_P v < 0$, so G is concave in P_{top} and its maximum is at the P_{top} for which the placement exactly fills the box. The reference energy is therefore $E_* = E_G - g z_0 M_{\mathrm{tot}}$ with

$$E_G = \min_x \max_{P_{\mathrm{top}}} G(x, P_{\mathrm{top}}) = \max_{P_{\mathrm{top}}} \min_x G(x, P_{\mathrm{top}}), \tag{15}$$

where the exchange holds because G is linear in the placement x, which ranges over a compact convex set, and concave in P_{top} (Sion's minimax theorem). G is the discrete background energy of the energy notes, restricted to hydrostatic profiles, and the maximum over P_{top} is their statement that $\int \Pi\,\mathrm{d}m$ is a minimum with respect to P_*. This gives the **algorithm for FC2D**:

1. outer loop: a one-dimensional maximisation over P_{top}, by Newton or secant iteration on the volume residual $R(P_{\mathrm{top}}) = \sum x_{nk} v_{nk} - L_x L_z$, with $R' = -\sum x_{nk} v_{nk}^2/c_{s,nk}^2$ (c_s the sound speed);
2. inner problem: the rearrangement problem (8) with the costs (13) at the current P_{top}.

Stop on $|R|/(L_x L_z)$ below a tolerance (e.g. 10^{-13}), not after a fixed number of iterations. If the placement is independent of P_{top} (the separable case of §2, solved by a sort) the inner problem is solved once. Otherwise, warm-start each inner solve from the previous placement: near convergence it rarely changes. This is the discrete analogue of the free-surface optimal transport problem of Egan et al. (2025), where the surface pressure plays the role of P_{top}.

Geometry of the solution. Given the converged placement and P_{top}, slot k has volume $V_k = \sum_n x_{nk} v_{nk}$, thickness $\Delta z_k = V_k/L_x$, and heights stacked from the bottom. The edge pressures are $P_{\mathrm{top}} + gkM_{\mathrm{tot}}/(KL_x)$, and the slot density $\rho_{*,k} = M_k/V_k$ is exactly $-\Delta P_*/(g\,\Delta z_k)$ across the slot, the discrete form of (3). Placement heights $\hat{z}_n$ and the sub-slot refinement are as for AN2D.

Initial guess for P_{top}. The mean of the top row of the actual pressure, or the value from the previous snapshot.

3.5 Dual potentials: the reference pressure

The dual of (8) has a potential u_n per parcel and ψ_k per slot, with

$$c_{nk} - u_n - \psi_k \geq 0 \quad \forall n, k, \qquad \text{with equality wherever } x_{nk} > 0. \tag{16}$$

So every slot that holds parcel n minimises $c_{nk} - \psi_k$ over k, and u_n is that minimum. In §1, z_* minimises $\Sigma(\vartheta, q, z, P_*(z,t))$. Comparing the two:

- **AN2D**: $\Sigma(\vartheta_n, q_n, z_k, P_{*,k}) = c_{nk} + \pi_{*,k}$ by (1), so

$$\pi_{*,k} = -\psi_k + \text{const}, \qquad P_{*,k} = P_{R,k} + \rho_{R,k}\,\pi_{*,k}. \tag{17}$$

- **FC2D**: $\Sigma(\vartheta_n, q_n, z_k, P_{*,k}) = c_{nk} + \Phi(z_k)$, so $\psi_k = -\Phi(z_k) + \text{const}$. The duals give the heights of the slot centres, which agree with the geometry of §3.4 to $O(\Delta z)$. The pressures are $P_{*,k} = P_k(P_{\text{top}})$.

In both models $u_n = \Sigma(\vartheta_n, q_n, z_*, P_*(z_*, t))$, up to the same constant: the reference static energy in (2). Dual feasibility with complementary slackness is the discrete LNB condition together with the maximal APE condition, in global form. And ψ is the multiplier of the slot constraints $\sum_n x_{nk} = M_k$, the discrete volume constraint, just as P_* is the Lagrange multiplier of the volume constraint in the energy notes. For a parcel split between slots k and k', equality for both slots gives $\psi_k - \psi_{k'} = c_{nk} - c_{nk'}$. In AN2D this is $\pi_{*,k} - \pi_{*,k'} = c_{nk'} - c_{nk}$, the discrete form of $\partial_z \pi_* = g(\rho_R v_* - 1)$ with v_* the specific volume of the parcel that straddles the interface. Where no parcel is split across an interface (a degenerate placement), the step of ψ there is only bracketed by the cost differences of the parcels on either side. This is the discrete counterpart of a jump in v_*.

Weak duality and the AE. The dual objective

$$D(\psi) = \sum_k M_k \psi_k + \sum_n m_n \min_k (c_{nk} - \psi_k) \tag{18}$$

is the background energy of the energy notes for the profile ψ, with every parcel at the z_* that maximises its Π. In AN2D this holds up to the constant $-\sum_k M_x P_{R,k} \Delta x \Delta z$; in FC2D, $D - P_{\text{top}} L_x L_z - g z_0 M_{\text{tot}}$ at the P_{top} of the costs. Weak duality gives $D(\psi) \leq \min_x \sum x_{nk} c_{nk}$ for *every* ψ (and every P_{top}). So for any reference profile, Lorenz or not, $\int \Pi \, \mathrm{d}m$ with the global minimiser z_* is at least $E_P - E_*$ of the Lorenz state, with equality for the optimal duals (cf. Tailleux 2013 for a Boussinesq fluid). In FC2D this holds up to the discretisation mismatch of §13. In AN2D with one slot per row it is exact: with z_* restricted to the row centres,

$$\sum_n m_n \Pi_n = E_P - \left(D(\psi) - \sum_k M_x P_{R,k} \Delta x \Delta z \right), \tag{19}$$

because the actual state also has mass M_k in row k. With the optimal duals this is $E_P - E_*$ to round-off. Refining z_* between the rows (§8.2) can only increase it.

4 Solver 1: sorting

Exact iff (6) **holds** (here: $\gamma = 0$). Cost $O(N \log N)$.

1. Sort key: the specific volume at a fixed key pressure, $\nu_n = v(P_{\text{key}}, \vartheta_n, q_n)$. When $\gamma = 0$ any P_{key} gives the same order. When $\gamma \neq 0$, choose the mid-depth pressure (P_R at $z = L_z/2$ for AN2D, the box mean of P for FC2D), since it minimises the largest pressure offset.

2. Order the parcels by decreasing ν (lightest first), i.e. the top of the column first.

3. Fill the slots from the top in that order, splitting a parcel across a slot boundary when needed. This gives x_{nk}.

4. FC2D: the placement does not change with P_{top} when the order is exact; run the outer Newton loop of §3.4 once on this placement. With $\gamma \neq 0$, the order can in principle be refreshed at the final pressures, which is not a minimisation; prefer solvers 2 or 3 if the sort is found to be inadequate.

At parcel resolution ($K = N$, a slot per parcel in sort order) this is the construction of Winters et al. (1995) and of the earlier `FCSCEnergetics.jl`.

Duals and certificate. The positive entries of the sorted placement form a staircase. Solve $u_n + \psi_k = c_{nk}$ along it from the top, in $O(N + K)$; where no parcel is split across a slot boundary, complete the staircase with a zero entry. Then check (16) for all NK parcel–slot pairs (an $O(NK)$ pass). If it holds, the sorted placement is the exact optimum, not only 2-optimal, and ψ is the reference profile of §3.5. If it fails, the violating pairs show where to repair (§5.3) or which pairs enter the transportation simplex (§6).

Pros. Fast and simple, with no dependency. It is the baseline against which every other solver is tested: with $\gamma = 0$ all solvers must agree with it to round-off.

Cons. Approximate when $\gamma \neq 0$. The certificate, or the duality gap of §5.3, measures the error.

5 Solver 2: greedy filling with repair

Approximate for any equation of state, with no dependency. Each method below evaluates the specific volume of the candidates at the pressure of the level being filled, so it accounts for thermobaricity.

5.1 Top-down filling

For $k = 1, \ldots, K$ from the top: evaluate $v(P_k, \vartheta_n, q_n)$ for the parcels not yet fully placed, and fill slot k with the largest values (lightest first), splitting the last parcel. Cost: $O(NK)$ evaluations of v (about 4×10^6 at 256×128, $K = 128$). This is the "greedy" algorithm of Huang (2005) and Wong et al. (2016), and the discrete counterpart of the top-down construction of Saenz et al. (2015). *Bottom-up* is the same from $k = K$ upwards, taking the smallest v first. Harris & Tailleux (2018) found top-down to be the least reliable heuristic for moist air, and bottom-up to systematically miss part of $E_P - E_*$ there.

5.2 Divide and conquer

Stansifer et al. (2017). On a slot range $[k_1, k_2]$ with its set of parcel mass: split it at the slot boundary k_m that halves the mass, evaluate v at the boundary pressure, and send the lightest parcels (by mass, splitting one parcel at the cut) to $[k_1, k_m]$ and the rest to $[k_m + 1, k_2]$. Recurse until single slots. Cost: $O(N \log K)$ evaluations. This was the best speed/accuracy compromise in Harris & Tailleux (2018), but it still had $> 50\%$ error in 11% of their cases and occasionally gave a negative $E_P - E_*$, so the repair below is not optional.

5.3 Repair (pairwise exchanges, 2-opt) and the energy check

An exchange of mass $\delta > 0$ between parcel n in slot k and parcel l in slot k' changes the cost by

$$\delta \, (c_{nk'} + c_{lk} - c_{nk} - c_{lk'}) \, . \tag{20}$$

Whenever the bracket is negative, exchange $\delta = \min(x_{nk}, x_{lk'})$. Repeat over all pairs of slots with $|k - k'| \leq B$ (a band; thermobaric misorderings are local, so a small B such as 2–4 should suffice; check by increasing B) until no improving exchange is found. The result is a local minimum, not necessarily the global one (Hieronymus & Nycander 2015 give examples where 2-opt fails). For FC2D the repair runs inside the outer P_{top} loop.

Energy check (mandatory for every approximate solver): the reference energy must not exceed the energy of the actual state, $E_* \leq E_P$. For AN2D with one slot per row the actual state is a feasible placement with exactly the same discrete energy, so a violation is a solver failure and must raise an error. For FC2D (and AN2D with $s > 1$) the two energies are computed on different grids (§13), so allow a tolerance set by the discretisation.

Duality gap: any feasible placement x gives $E[x] \geq E_*$, and any ψ gives $D(\psi) \leq E_*$ (up to the constants of §3.5). With ψ from the placement (the staircase of §4) or from the slot means (§3.3), $E[x] - D(\psi)$ bounds the error of the approximate solver at the cost of one $O(NK)$ pass.

Pros. No dependency; handles thermobaricity; near-exact in practice after repair.

Cons. No optimality guarantee; heuristics without repair can give a negative $E_P - E_*$; slower than the sort.

6 Solver 3: exact rearrangement

Solves (8) exactly for any equation of state. Its outputs are the placement x and the dual potentials (u_n, ψ_k), which certify optimality by (16) and give the reference profile (§3.5).

6.1 Transportation simplex with a dual certificate (implemented)

Solver: a transportation simplex written in Julia, generic in the floating-point type F (external LP solvers such as HiGHS work in Float64 only, which the precision requirement of §13 excludes). Its network has $N + K$ nodes, the parcels and the slots, with an arc from each parcel to each slot that carries x_{nk}; a basis is a spanning tree of $N + K - 1$ arcs.

1. **Start**: the sorted placement (§4), whose entries form a staircase, completed into a tree: at a degenerate boundary between slots k and $k + 1$ (no parcel split across it), the first parcel b of slot $k + 1$ gets an entry of zero mass in slot k.

2. **Duals**: $\psi_1 = 0$ and $u_n + \psi_k = c_{nk}$ along the tree, by a breadth-first walk from the first slot, in $O(N + K)$ with the cost of each tree arc kept.

3. **Pricing**: a pass over all NK pairs (the certificate pass, threaded and vectorised) gives the least reduced cost of each parcel and its slot. Each parcel with a negative one is a candidate; the candidates are taken most negative first, priced again with the current duals (one cost each), and enter the tree if still negative. The pass repeats until it finds no negative reduced cost: the last pass is the certificate.

4. **Pivot**: the entering pair closes a cycle in the tree, around which a mass θ moves ($+\theta$ on the arcs oriented as the entering one, $-\theta$ on the others), the least mass of the arcs that lose it; one of these leaves.

The tree is rooted at the first slot and kept *strongly feasible* (Cunningham 1976): every arc of zero mass goes from a parcel up to its parent slot, which the completed staircase satisfies. The leaving arc is the last arc that blocks θ met going round the cycle from its apex in the direction of the entering arc; with this rule a pivot that moves no mass ($\theta = 0$, frequent from the degenerate staircase) still makes progress, and the method cannot cycle. The tree and the duals are rebuilt after each pivot, in $O(N+K)$. The pivots give the decrease of the cost exactly, $-\sum \theta\,(c_{nk} - u_n - \psi_k)$: the amount by which the sort underestimates $E_P - E_*$.

FC2D: the costs depend on P_{top}, so the two steps of §3.4 alternate from the sorted placement: P_{top} at fixed placement (Newton), then the simplex at fixed P_{top} from the previous tree, until a solve moves no mass.

Cost: the number of passes grows with how far the sort is from the optimum: a few where the sort fails for a few parcels, about K where every parcel is misplaced (two density-compensated families mixed in every row): each pass moves a misplaced parcel about one slot nearer its level. Each pass is NK evaluations of the cost, which dominate. The method cannot cycle, so the cap on the passes, $10K$ by default (decided with the user, 2026-10-27), only guards against round-off; it scales with the slots as the passes do, rather than with the parcels, which would make a failure take N/K times longer to report.

Measured (2026-10-27, `tools/exact_rearrangement_study.jl`):

- thermohaline lock exchange ($\gamma = 10^{-7}\,\mathrm{Pa}^{-1}$): the sort fails the certificate for at most 3 nearly tied parcels, which 2 or 3 pivots fix; the exact rearrangement changes $E_P - E_*$ by at most 9×10^{-10} of it and P_* by at most $2 \times 10^{-8}\,\mathrm{Pa}$, with the same roughness in time: the sort is the Lorenz state to round-off there. Cost: that of the sort when it has no violation (one pass), 2.6 to 3.5 times with a few;

- two density-compensated families mixed in every row ($\gamma = 1.3 \times 10^{-4}\,\mathrm{Pa}^{-1}$): every parcel is misplaced by the sort; 315, 1114 and 7405 pivots in 21, 38 and 104 passes at $K = 20$, 40, 80 (0.46 s at $N = 1600$);

- Float32 sees no reordering on the active state of the tests for γ up to $10^{-3}\,\mathrm{Pa}^{-1}$, where Float64 makes 95 pivots that move mass: the reduced costs are below its tolerance $16(K+1)\,\epsilon\,gL_z$. The exact rearrangement needs Float64.

Not done (decided with the user, 2026-10-27, to be tried if they prove the fastest): a warm start from the previous snapshot's placement (the parcels are the model cells, so after advection the previous placement has no direct relation to the new state), pricing on a band $|k - k_n| \le B$ around the sorted slot before the full pass, and incremental updates of the tree (thread indices) in place of the rebuild. For equal-mass parcels and $K = N$ the problem is the linear assignment problem (Hungarian or Jonker–Volgenant), but that is $O(N^3)$ and not needed here. Su & Ingersoll (2016) report min-cost flow 10^3–10^5 times faster than linear assignment on the same problem, by exploiting $K \ll N$, which is the structure used here.

6.2 Alternative: dual ascent

With the transportation simplex this is not needed, but it is another route and it makes the link with §1 explicit. The dual of (8), after eliminating u, is the maximisation of the concave piecewise-linear $D(\psi)$ (18) in only K variables, with supergradient $M_k -$ (mass of the parcels whose minimiser is k). It is the discrete form of the variational principle of §1: for a given profile every parcel takes the level that maximises its Π, and the profile is adjusted until the volume constraint holds; the supergradient is the volume residual of each slot. It is also the discrete counterpart of the semi-discrete optimal transport (Laguerre-cell) formulation of Egan et al. (2025). It can be maximised by a bundle or subgradient method, or by entropic smoothing (log-domain Sinkhorn with decreasing regularisation). The placement

is read off from the minimisers. Parcels with ties must be split by a small flow among the tied slots. This needs more numerical care than the LP route.

Pros. The true Lorenz state, with a certificate and the reference pressure; handles unequal parcel masses and any equation of state; the band keeps it close to the cost of the sort.

Cons. Our own code to maintain. It is slower than the sort, and for FC2D it is repeated for each P_{top} iteration (warm-started).

7 Solver 4: histogram (PDF) reduction

Tseng & Ferziger (2001), generalised to two tracers by Saenz et al. (2015). This is a *reduction* rather than a solver: it replaces the N parcels by N_b bins before applying one of solvers 1–3.

1. Bin the parcels in (ϑ, q) (or in the sort key ν if separable). For each nonempty bin b store its mass $m_b = \sum_{n\in b} m_n$ and representative properties: the mass-weighted means $\bar{\vartheta}_b, \bar{q}_b$.
2. Treat the bins as parcels and solve (8) with solver 1, 2 or 3.
3. Give the parcels their reference height by the LNB equation against the resulting reference profile (§8.2), which gives a continuous z_* for each parcel rather than one per bin.

In the separable case the profile follows directly from cumulative mass: a bin at sort key ν sits at the height where the mass of lighter fluid above it equals the mass of all bins with larger ν (Tseng & Ferziger 2001, eq. 6, with mass in place of volume).

Pros. $O(N + N_b)$; a smooth profile with a resolution set by the bins rather than the grid; the bin counts are reusable across diagnostics.

Cons. Averaging (ϑ, q) inside a bin with a nonlinear equation of state biases the energy (cabbeling), so bins must be fine. A resting, stably stratified state no longer gives zero available energy exactly. At $N \approx 3 \times 10^4$ the saving over the sort is small; the benefit is for much larger grids or long time series.

8 Reference heights: the LNB equation

Given a reference profile, Lorenz (§3) or horizontal mean (§9), each parcel gets its z_* by minimising $\Sigma(\vartheta, q, z, P_*(z,t))$, as in §1.

8.1 Profile representation

A profile is known at K levels (slot edges or centres, or model rows) and must be interpolated to solve the LNB equation. Interpolate the *pressures* and derive the specific volumes from the interpolant. Then Π, the LNB residual, v_* and N_*^2 all come from one profile, and the z_* found below is an exact minimiser of the Σ that enters Π:

- **AN2D**: interpolate P_R and π_* between the row centres with C^1 cubic Hermite interpolants (for P_R with the node slopes $-g\rho_{R,k}$, so that it matches the model's P_R and ρ_R at the centres). Set $\rho_R(z) = -P_R'(z)/g$ and $v_*(z) = \left(\pi_*'(z)/g + 1\right)/\rho_R(z)$ by (3);
- **FC2D**: interpolate P_* in height through the slot-edge pressures (Lorenz) or the row values (horizontal mean) with a monotone C^1 cubic, and set $\rho_*(z) = -P_*'(z)/g$, $v_* = 1/\rho_*$. Do not interpolate $\rho_{*,k}$ independently.

In FC2D the levels include the walls. In AN2D the levels are the row centres, half a row from the walls, and between the end rows and the walls v_* follows the compression of the parcels of the end slot e, $v_* = v_e + b_e\,(P_R(z) - P_R(z_e))$, with b_e the mass-weighted mean of their $\partial v/\partial P$ at $P_R(z_e)$. A constant v_* there lowers the static energy of these parcels towards both walls by $\frac{1}{2}|b_e|(g\rho_R\Delta z/2)^2$; in a neutral fluid, where Σ is otherwise flat along the profile, a wall then becomes the reference height of every parcel, each with Π of that size (10% of the available energy of the rising bubble of `examples/energy_budget`). With v_* following the end slot's parcels, a neutral or resting fluid has Π at round-off when ρ_R is uniform, and $O(\Delta z^4)$ from the cubic interpolant of π_* otherwise. A piecewise-linear interpolant also works with the global minimum of §8.2, but it gives a piecewise-constant v_*, and the minimum may then sit at a level, where the residual changes sign without vanishing.

8.2 The LNB equation and the choice of root

For a parcel (ϑ, q), write $S(z) = \Sigma(\vartheta, q, z, P_*(z,t))$ for its static energy along the profile, (1). Its reference height is the global minimiser

$$z_* = \arg\min_{0\le z\le L_z} S(z) \tag{21}$$

(§1). At an interior minimum $r(z_*) = 0$, with the *LNB residual*

$$r(z) = -\frac{\mathrm{d}S}{\mathrm{d}z} = \begin{cases} g\,\rho_R(z)\,(v(P_R(z),\vartheta,q) - v_*(z)) & \text{AN2D,} \\ g\,(\rho_*(z)\,v(P_*(z),\vartheta,q) - 1) & \text{FC2D,} \end{cases} \tag{22}$$

and there $N_*^2 = -\mathrm{d}r/\mathrm{d}z \ge 0$. $r > 0$ means the parcel is lighter than the reference at z. N_*^2 depends on the parcel through (ϑ, q) and not only on height (Tailleux 2013, Appendix A).

1. **Coarse search:** evaluate S at the K levels of the profile and at the walls (cheap: $K \approx 128$ evaluations per parcel) and list its local minima. At the levels, $S = c_{nk} + \pi_{*,k}$ (AN2D) or $h(P_{*,k}, \vartheta, q) + \Phi(z_k)$ (FC2D). For a Lorenz profile from the duals this is $c_{nk} - \psi_k$ up to a constant (§3.5), the same pass as the certificate of §6.

2. **Refine** each local minimum whose value is within a tolerance of the smallest. On the intervals next to it, find the root of r with $\mathrm{d}r/\mathrm{d}z < 0$ by Brent's method on the interpolated profile, to a tolerance well below Δz. Keep the smallest refined S. A wall is the minimum if r has the outward sign there ($r \ge 0$ at the top, $r \le 0$ at the bottom).

3. **Record** N_*^2; the parcels at a wall, where $r(z_*) \ne 0$ (evaluate their budget terms with the general forms of the energy notes, $\Sigma_P(\vartheta, q, z_*, P_*(z_*,t))$ rather than $v_*(z_*,t)$); and the parcels with two minima within the tolerance, where z_* can jump between them.

Why the global minimum. It maximises the parcel's Π at fixed P_* (§1), so it depends on (ϑ, q, t) only, not on the parcel's actual height, and $\Pi_2 = S(z) - S(z_*) \ge 0$. $S(z_*)$ is continuous in (ϑ, q, t) even where z_* jumps between two minima. (Σ_ϑ and Σ_q at z_*, which enter the heating terms of the energy notes, do jump there.) For the Lorenz profile it reproduces the levels of the placement, and for any profile it gives the bound of §3.5. Tailleux (2013, §3.1) instead takes the root nearest to the parcel's height z: above z if $r(z) > 0$, below if $r(z) < 0$, or the wall if there is none. This is also positive definite, but then the reference height depends on z, and Π jumps when a parcel crosses a local maximum of S, so it is not used here. The two rules agree when S has a single local minimum.

Multiple minima arise for the Lorenz profile only through the pressure dependence of the parcel ordering ($\gamma \ne 0$; the overlap region of Saenz et al. 2015). For the horizontal mean they also arise where the mean profile is statically unstable, e.g. during overturning in Kelvin–Helmholtz billows. Count and record parcels with multiple minima and with a wall minimum.

8.3 Against the Lorenz profile

For the Lorenz profile, z_* agrees with the placement height $\hat{z}$ to the slot resolution. In slots that hold a mixture of parcels v_* is an average, so the two can differ by up to a slot thickness there. The budgets use z_*: it satisfies the LNB condition for the profile used in Π, so the $\partial_t z_*$ terms vanish (§1). The placement height $\hat{z}$ satisfies the volume constraint exactly but the LNB condition only to $O(\Delta z)$; for z_* it is the other way round, and the LNB condition is the one the local budgets need. The LNB z_* is also a single height for a parcel split between slots, so the fractions x_{nk} are not needed for Π.

9 Horizontal-mean reference state

This reference state is *not* a rearrangement of the fluid: its P_* does not satisfy the volume constraint (4). The local AE and its local budgets in the energy notes hold for any $P_*(z,t)$ with the z_* of §8.2. But the reference pressure tendency term R_Π no longer integrates to zero, and $\int \Pi \,\mathrm{d}m$ exceeds $E_P - E_*$ of the Lorenz state (§3.5; Tailleux 2013). It is cheap and needs no sort.

9.1 The profiles

The profile is specified by $P_*(z,t)$ alone, hydrostatic with the horizontal mean of the quantity in the model's hydrostatic balance. In FC2D this is the density, the usual choice in the literature (Tailleux 2013; Tailleux 2018, §4.2):

- **AN2D.** The thermodynamics stays at P_R. π_* is hydrostatic with the row mean $v_{*,k} = \frac{1}{M_x}\sum_i v(P_{R,k}, \vartheta_{ik}, q_{ik})$, by the rule of §3.3 with $s = 1$. This is the row mean of the buoyancy $b = g(\rho_R v - 1)$, which is linear in v, as the FC2D balance is linear in ρ.
- **FC2D.** P_* is hydrostatic with the row-mean density $\bar{\rho}_k = \frac{1}{M_x}\sum_i m^\rho_{ik}$ by the model's discrete rule $P_{*,k} - P_{*,k-1} = -g\Delta z\, \mathcal{I}_z \bar{\rho}$, anchored to the row-mean pressure at one level. The cells have equal volume, so this keeps the mass of each row, and $\rho_* = \bar{\rho}$.

In both, v_* is still derived from the interpolant of P_* (or π_*) of §8.1. Hermite node slopes from the row means ($-g\bar{\rho}_k$, or $g(\rho_{R,k} v_{*,k} - 1)$) make it equal to the row mean at the centres. The row mean of the pressure itself is not used. It contains the non-hydrostatic part of the mean pressure, so its implied density $-\partial_z P_*/g$ is not $\bar{\rho}$ and it does not keep the row masses.

10 Time derivative of the reference pressure

The reference pressure tendency terms R_{Π_1}, R_{Π_2} and R_Π of the energy notes need $\partial_t P_*(z,t)$ on the model rows and at each parcel's z_*, from the same interpolant as P_* (§8.1). By default it is computed from the diabatic tendencies of a single snapshot (§10.1); differences of successive snapshots (§10.2) serve as a check. Both hold for any equation of state: they use only $v(P,\vartheta,q)$ and its derivatives, and keep the compressibility v_P wherever it enters.

The global identity $\int R_\Pi \,\mathrm{d}V = 0$ of the energy notes uses only the volume constraint (4) at time t, with $f = \partial_t P_*$. It therefore holds however $\partial_t P_*$ is estimated, to the accuracy with which z_* satisfies the volume constraint (§8). The estimate affects only the local distribution of R_Π and, in FC2D, the exchange $\int R_{\Pi_1} \,\mathrm{d}V$ between Π_1 and Π_2.

10.1 From the diabatic tendencies (default)

The Lorenz reference state depends only on the distribution of mass over (ϑ, q). Advection rearranges the fluid without changing that distribution, so P_* changes only through the diabatic

rates of the fluid elements (Winters et al. 1995; Winters & D'Asaro 1996),

$$\dot{\vartheta} = \frac{D\vartheta}{Dt} = \frac{\sigma_\vartheta - \nabla\cdot\mathbf{J}_\vartheta}{\rho}, \qquad \dot{q} = \frac{Dq}{Dt} = -\frac{\nabla\cdot\mathbf{J}_q}{\rho}, \tag{23}$$

which the model provides (`conservative_variable_production`, `div_consflux`, `div_qflux`; $\rho = \rho_R$ in AN2D). For a parcel at pressure P, let

$$\dot{v} = v_\vartheta\,\dot{\vartheta} + v_q\,\dot{q} \tag{24}$$

be the rate of change of its specific volume at fixed pressure, with $v_\vartheta = \partial_P h_\vartheta$ and $v_q = \partial_P h_q$ (or a directional derivative of $v(P,\vartheta,q)$).

FC2D. In the mass coordinate m of §3.4 (the mass above a level), $P_*(m,t) = P_{\mathrm{top}}(t) + gm/L_x$. Let $v_*(m,t)$ be the reference specific volume and $m_*(\vartheta,q,t)$ the reference mass coordinate of a parcel. Follow a fluid element, whose mass is fixed, and differentiate the LNB condition $v(P_*(m_*,t),\vartheta,q) = v_*(m_*,t)$ along it. With $\partial_m = -(v_*/L_x)\,\partial_z$ and $N_*^2 = g\rho_*^2\,(v_*\partial_z v_* + g v_P)$ (energy notes, "Maximal APE condition"),

$$\frac{N_*^2}{g\rho_*^2 L_x}\,\frac{Dm_*}{Dt} = \partial_t v_*|_m - \dot{v} - v_P\,\partial_t P_{\mathrm{top}}, \tag{25}$$

with $\dot{v}$, v_P and N_*^2 those of the parcel at its reference level. The mass above the level m is m itself at all times, so the mass-weighted mean of Dm_*/Dt over the parcels at a level vanishes. Hence

$$\partial_t v_*|_m = \frac{\left\langle (\dot{v} + v_P\,\partial_t P_{\mathrm{top}})\,/N_*^2\right\rangle_m}{\left\langle 1/N_*^2\right\rangle_m}, \tag{26}$$

where $\langle\cdot\rangle_m$ is the mass integral over the parcels whose reference level is m. The box stays full, $\int_0^{M_{\mathrm{tot}}} \partial_t v_*|_m\,\mathrm{d}m = 0$, which fixes $\partial_t P_{\mathrm{top}}$: a heated column expands, and the lid compresses it back. With $z(m,t) = L_x^{-1}\int_m^{M_{\mathrm{tot}}} v_*\,\mathrm{d}m'$, the tendency at fixed height is

$$\partial_t P_*|_z = \partial_t P_{\mathrm{top}} + g\rho_*\,\partial_t z|_m, \qquad \partial_t z|_m = \frac{1}{L_x}\int_m^{M_{\mathrm{tot}}} \partial_t v_*|_{m'}\,\mathrm{d}m'. \tag{27}$$

AN2D. The mass per unit height is fixed, so the levels do not move and there is no lid. The same steps with $v(P_R(z_*),\vartheta,q) = v_*(z_*,t)$ give $(N_*^2/g\rho_R)\,Dz_*/Dt = \dot{v} - \partial_t v_*$, hence

$$\partial_t v_*(z,t) = \frac{\left\langle \dot{v}/N_*^2\right\rangle_z}{\left\langle 1/N_*^2\right\rangle_z}, \qquad \frac{\partial}{\partial z}\,(\partial_t \pi_*) = g\rho_R\,\partial_t v_*, \qquad \partial_t P_* = \rho_R\,\partial_t \pi_*, \tag{28}$$

with $\dot{v}$ and $N_*^2 = g\rho_R\,(\partial_z v_* + g\rho_R v_P)$ at $(P_R(z_*),\vartheta,q)$. The constant in $\partial_t\pi_*$ is free, since R_Π does not depend on it.

The weights. The reference state follows the N_*^{-2}-weighted mean of the parcels at a level. A parcel that is weakly stratified at its level moves further for a given change of v, so it displaces more of its neighbours. The weights differ between the parcels of a level only through v_P, i.e. through the pressure dependence of v (thermobaricity), and they differ most where N_*^2 is small. The plain mean is exact only when v_P is the same for all of them, e.g. for (6). It is also what a derivative at fixed placement gives, since a fixed placement does not see the parcels reorder within a level. So the weights must be kept for a general equation of state (§14, test 10).

Discrete form. The parcels at a level are those of a slot, with the weights of the placement. For parcel n in slot k, evaluate $\dot{v}_{nk}$ and $v_{P,nk}$ at the slot pressure ($P_{R,k}$ in AN2D, P_k in FC2D), and

$$N^2_{nk} = g\rho_k\,(\partial_z v_* + g\rho_k\, v_{P,nk})\,, \qquad w_{nk} = \frac{x_{nk}}{N^2_{nk}}, \qquad \langle a\rangle_k = \frac{\sum_n w_{nk}\, a_{nk}}{\sum_n w_{nk}}, \tag{29}$$

with $\rho_k = \rho_{R,k}$ (AN2D) or $\rho_{*,k}$ (FC2D), and $\partial_z v_*$ at z_k from the profile of §8.1.

- **AN2D**: $\partial_t v_{*,k} = \langle \dot{v}\rangle_k$, and $\partial_t \pi_*$ by integrating (28) between the slot centres with the rule used for π_* (§3.3): the node slopes $g\rho_{R,k}\partial_t v_{*,k}$, the values by the trapezoidal rule in P_R, $\partial_t\pi_*(z_c) - \partial_t\pi_*(z_{c-1}) = -\frac{1}{2}(\partial_t v_{*,c-1} + \partial_t v_{*,c})(P_R(z_c) - P_R(z_{c-1}))$, and a zero mean over the mass of the rows, as for π_*. Between the end rows and the walls, where $v_* = v_e + b_e(P_R - P_R(z_e))$ (§8.1), only $\partial_t v_e$ is kept: $\partial_t b_e$ needs the mixed derivatives $\partial^2 v/\partial P\partial\vartheta$, $\partial^2 v/\partial P\partial q$, which the equation of state does not provide. The neglected term of $\partial_t\pi_*$, $-\partial_t b_e\,\Delta P^2/2$, is at most 2.6×10^{-3} of $\max|\partial_t\pi_*|$ for a thermobaric coefficient $\gamma = 3\times10^{-4}\,\mathrm{Pa}^{-1}$ at $M_z = 16$ (and 10^{-6} for the γ of the tests), and it changes $\int R_{\Pi_2}\,\mathrm{d}V$ by about 10^{-4} of $\int|R_{\Pi_2}|\,\mathrm{d}V$ (`tools/reference_tendency_study.jl`);

- **FC2D**:

$$\partial_t P_{\mathrm{top}} = -\frac{\sum_k M_k\langle\dot{v}\rangle_k}{\sum_k M_k\langle v_P\rangle_k}, \qquad \partial_t v_{*,k} = \langle\dot{v}\rangle_k + \langle v_P\rangle_k\,\partial_t P_{\mathrm{top}}. \tag{30}$$

  The slot edges move at $\partial_t z_e = L_x^{-1}\sum_{k\ \text{below}\ e} M_k\,\partial_t v_{*,k}$, which vanishes at both walls, and the edge pressures at $\partial_t P_{\mathrm{top}}$. $\partial_t P_*$ at fixed height is the derivative of the interpolant of §8.1 with respect to these node motions; at the edges this is (27). It is computed with dual numbers, value $+\,\varepsilon\times$ tendency, for the heights, pressures and secants of the edges, through the same monotone slopes and Hermite cubic as P_*: the derivative of every formula of the interpolant, on the same branch. The slot masses M_k are those of the rows of the state; the tendency holds them fixed, although advection changes them.

If $N^2_{nk} \le 0$ for a parcel of a slot (a marginally or discretely unstable level), the weights are undefined there: use $w_{nk} = x_{nk}$ in that slot and flag it. The tendency is that of the slot-level profile, so it matches the dual profile of §3.5 to $O(\Delta z)$. With it, $\int R_\Pi\,\mathrm{d}V$ vanishes to first order in the slot size, as the reference heights of the parcels of a mixed slot differ from their placement by up to a slot thickness: $|\int R_\Pi|/\int|R_\Pi|$ is 2×10^{-2} to 2×10^{-3} from $M_z = 8$ to 64 in the tests.

Horizontal-mean reference state. Its profile is a row average, not a function of the distribution of (ϑ, q), so it also changes under advection. Differentiate its definition (§9.1) with the model's full tendencies. In AN2D, $\partial_t v_{*,k}$ is the row mean of $v_\vartheta\partial_t\vartheta + v_q\partial_t q$ at $P_{R,k}$. In FC2D, $\partial_t\bar{\rho}_k$ is the row mean of the mass tendency, $\partial_t P_*$ is hydrostatic with it by the same discrete rule, and the anchor moves with the row mean of $\partial_t P = (\partial_t v - v_\vartheta\partial_t\vartheta - v_q\partial_t q)/v_P$ at its level.

10.2 From differences of successive snapshots (check)

Compute the reference state at each snapshot, warm-starting the solver from the previous placement. Evaluate P_* on the model rows, take centred differences in time, and evaluate the result at z_* with the interpolant of the middle snapshot.

- In AN2D the constants in π_* need not agree between snapshots, since R_Π is unchanged under $\pi_* \to \pi_* + c(t)$.

- Where the placement is degenerate the duals are not unique (§3.5), and a solver may return a different value at each snapshot. Difference the slot-mean profile there, or make the choice of dual reproducible.

- This is the actual time derivative of the diagnosed reference state, including the effect of numerical mixing. A budget of Π differenced between the same snapshots closes against it, to $O(\Delta t^2)$.
- The cell-based reference state is not exactly invariant under discrete advection: as features cross cells, the histogram of cell values changes slightly, at about $|\mathbf{u}|/\Delta x$. Use a snapshot interval $\Delta t \gg \Delta x/|\mathbf{u}|$, or average, so that this does not alias into $\partial_t P_*$. The interval must still be short compared with the diabatic evolution of P_*.

Using both. The estimate from the diabatic tendencies excludes, by construction, any change of the reference state caused by the discretisation of advection; the snapshot differences include it. Their difference therefore measures the spurious (numerical) change of the reference state. This is the local counterpart of using the background potential energy to measure spurious mixing (Ilıcak et al. 2012), and is useful for comparing the schemes of the models.

11 Outputs

Each solver returns, for one snapshot, what the energy notes need (§1):

- the **reference profile**, in the representation of §8.1. AN2D: $\pi_{*,k}$ on the slot centres (so $P_* = P_R + \rho_R \pi_*$). FC2D: the slot edges and centres z_k, $P_{*,k}$, $\rho_{*,k} = M_k/V_k$ and P_{top}. Both also return P_* and v_* at the model rows z_j and faces, from the same interpolant;
- per parcel: z_* (the global minimiser of §8.2); the reference static energy $\Sigma(\vartheta, q, z_*, P_*(z_*, t))$; the pressure at which the thermodynamics is evaluated there ($P_*(z_*, t)$ in FC2D, $P_R(z_*)$ in AN2D, with $\pi_*(z_*, t)$); N_*^2; the placement height $\hat{z}$; and flags for split, wall and multiple-minimum parcels;
- the **reference pressure tendency** $\partial_t P_*$ on the model rows and at each z_*, from the diabatic tendencies (§10.1), with $\partial_t v_{*,k}$, $\partial_t P_{\mathrm{top}}$ (FC2D) and the slots where the weights were replaced. Given a time series, also from differences of successive snapshots (§10.2), as a check;
- the **potential energies** E_P of the actual state and E_* of the reference state, in the same discretisation (AN2D: $E[x]$; FC2D: (14)), and $\int \Pi \, \mathrm{d}m$ from (2) as a check (§3.5);
- for the exact rearrangement, the duals and the certificate status; for the iterative ones, the iteration counts and final residuals.

In AN2D the constant in π_* is free at each time: Π, z_* and the reference pressure tendency term R_Π of the energy notes are unchanged under $\pi_* \to \pi_* + c(t)$. In FC2D, P_* is absolute (it is fixed by P_{top}). FC2D energies and available energies carry the $O(\Delta z^2)$ grid mismatch accepted in §13.

12 Summary of the methods

Method	Exact?	Cost	Use
Sort	iff $\gamma = 0$	$O(N \log N)$	baseline, default when certified (§4)
Greedy + repair	no (local minimum)	$O(NK)$	fallback if the exact solver is too slow
Exact rearrangement	yes, certified	simplex on a band	ground truth; validation of 1, 2, 4; P_* from the duals
Histogram	to bin resolution	$O(N + N_b)$	large grids, long time series
Mean + LNB	not a rearrangement	$O(NM_z)$	cheap, local reference; $\int \Pi \, \mathrm{d}m$ above Lorenz

Order of implementation: the sort with its certificate and the LNB (§8) first, with the mean profile (simple and cheap); then the exact rearrangement, as the reference solver and to measure the error of the sort with $\gamma \neq 0$; then greedy+repair and the histogram only if the exact solver is too slow for the intended use.

13 Decisions

Settled with the user.

- **Horizontal-mean profile**: hydrostatic with the row mean of the density (FC2D) or of the specific volume at P_R (AN2D), §9.1.
- **Exact solver**: a transportation simplex written in Julia, generic in the floating-point type, started from the sorted placement at each snapshot (§6). External LP solvers (HiGHS) are excluded because they work in Float64 only. Implemented (2026-10-27) with pricing by full passes over the pairs and a strongly feasible tree; the sort and the exact rearrangement stay separate methods, to be compared, and the examples use the exact one. The warm start from the previous snapshot, banded pricing and incremental tree updates are deferred.
- **Slot resolution** K: an input set by the user (§3.3, §3.4).
- **Precision**: any floating-point type F, chosen by the user (Float128 benchmarks are planned), so no step of the computation may be fixed to Float64.
- **Lorenz profile**: from the slot-mean specific volumes, in both models (§3.3); the duals certify the placement only. (First decided as the duals when the solver provides them; changed on 2026-10-27, as the dual profile made the budgets rough.)
- **Reference height for the budgets**: the LNB z_*, not the placement height $\hat{z}$ (§8).
- **Choice of root**: the global minimiser (§8.2), not the nearest root of Tailleux (2013).
- **AN2D profile beyond the end rows**: v_* follows the compression of the parcels of the end slot (§8.1), not a constant v_*, which gave every parcel of a neutral fluid a spurious $\Pi = O(\Delta z^2)$ at a wall (2026-10-26).
- **Reference pressure tendency**: from the diabatic tendencies (N_*^{-2} weights, any equation of state) by default, with differences of successive snapshots as a check (§10).
- **FC2D $\partial_t P_*$ at fixed height**: the derivative of the monotone cubic P_* with respect to the motion of its nodes, by dual numbers (§10.1), 2026-10-27. Three methods were compared: dual numbers and a hand-written tangent of the interpolant agreed to round-off and matched differences of the reference state to their floor, at the same cost; the formula (27) applied between the edges, with $\partial_t z|_m$ linear there, is not the derivative of the P_* of Π (off by 0.4 to 0.7% at $M_z = 64$ to 16). The dual numbers reuse the code of the interpolant.
- **AN2D beyond the end rows**: $\partial_t b_e$ is neglected (§10.1), its size measured, 2026-10-27.
- **Test of the weights under thermobaricity** (test 10b) is done with the exact rearrangement (milestone 2): the sorted placement reorders the parcels by its sort key, not by their level of neutral buoyancy. Done (2026-10-27): with two density-compensated families mixed in every row, of which one is heated, the N_*^{-2}-weighted mean is 26%, 8.5% and 5.7% from the differences of the exact rearrangement at $K = 20$, 40, 80, and the plain mean 35%.

Accepted limitation: the FC2D reference grid. In AN2D the reference column is built on the model's own rows (one slot per row), and the actual state is one of the candidate rearrangements, scored with exactly the model's energy. A fluid that is already at rest and stably stratified is then its own reference state, with zero available energy to round-off. FC2D cannot be set up this way. In a compressible column, the mass that fits in a row of fixed height depends on which parcels are put there, so the capacities of the rows are not known until the reference state is. The reference column is therefore built from layers of fixed mass, whose thicknesses come out of the calculation (§3.4). These layers do not line up with the model rows. Their energy is also computed by a different formula, (14), rather than the model's sum over cells with its discrete hydrostatic balance. Both are second-order accurate, but they do not give the same number: a fluid already in its Lorenz state gets an available energy $E_P - E_*$ of order Δz^2, of either sign. This decreases as the model grid is refined, but not by refining K alone, because part of it is the model's own discretisation error. Removing it would need slots on the model rows with unknown masses, a free-surface rearrangement problem that is harder to solve. **The $O(\Delta z^2)$ mismatch is accepted** (decision of the user). It is present in every FC2D result: the energy check of §5.3 uses a matching tolerance, test 1 of §14 measures it, and FC2D diagnostics should report it alongside the available energy.

14 Tests

1. **Resting stable state**, AN2D with one slot per row: a horizontally uniform, stably stratified state gives $x =$ identity, $\hat{z} = z$ and $E_* = E_P$ to round-off, for every solver. With the dual profile and z_* restricted to the row centres, $\Pi = 0$ to round-off for every parcel. For FC2D (and AN2D with $s > 1$) the same holds up to $O(\Delta z^2)$, converging as the model grid is refined.

2. **Separable equation of state** ($\gamma = 0$): sort, greedy+repair and exact rearrangement give the same E_* to round-off on random and Kelvin–Helmholtz snapshots, and the sort certificate holds.

3. **Thermobaric** ($\gamma \neq 0$): $E_{*,\mathrm{exact}} \leq E_{*,\mathrm{greedy+repair}} \leq E_{*,\mathrm{sort}}$ (the middle inequality holds when the repair starts from the sort). Report the relative gaps.

4. **Bounds**: $E_* \leq E_P$ (AN2D with $s = 1$ to round-off, otherwise within the discretisation tolerance). For any profile (Lorenz from any solver, horizontal mean) with z_* the global minimiser, $\int \Pi \, \mathrm{d}m \geq E_P - E_{*,\mathrm{exact}}$, with equality to round-off in AN2D ($s = 1$) for the optimal duals and z_* on the row centres.

5. **Conservation**: $\sum_k x_{nk} = m_n$, $\sum_n x_{nk} = M_k$; FC2D $|R(P_{\mathrm{top}})|$ below tolerance and $\sum_k \Delta z_k = L_z$.

6. **Permutation invariance**: shuffling the parcels (e.g. shifting the field periodically in x) leaves E_* and the profile unchanged.

7. **Gauge**: changing z_0 shifts E_* and E_P by the same constant; the reference heights and Π are unchanged. In AN2D, $\pi_* \to \pi_* + c$ leaves z_* and Π unchanged.

8. **LNB**: $r(z_*) = 0$ to tolerance at interior minima, with $N_*^2 \geq 0$; $\Pi \geq 0$ for every parcel ($\Pi_1 \geq 0$ by the concavity of Σ in P, $\Pi_2 \geq 0$ by construction). For the Lorenz profile, $z_* = \hat{z}$ up to the slot resolution in unmixed slots.

9. **Certificate**: the duals of the exact solver satisfy (16) for all parcel–slot pairs.

10. **Reference pressure tendency** from the diabatic tendencies:

(a) with v_P the same for all parcels of a level ($\gamma = 0$), it equals the finite-difference derivative of the reference state for a small diabatic step;

(b) with strong thermobaricity, e.g. two density-compensated families of parcels with different v_P, heated differently: the finite-difference derivative of the exact solver, which includes the reordering, matches the weighted mean, and not the plain one;

(c) FC2D: $\sum_k M_k \, \partial_t v_{*,k} = 0$, so the box stays full;

(d) with no diffusion, it gives $\partial_t P_* = 0$;

(e) $\int R_\Pi \, \mathrm{d}V = 0$ with this estimate and with the snapshot differences, to the slot resolution. The two estimates agree as the resolution and the snapshot interval are refined, up to the effect of numerical mixing, which is reported.

References

- Cunningham (1976), A network simplex method, Math. Programming 11, 105–116.
- Egan, Methven, Bourne & Cullen (2025), A definition of the background state of the atmosphere using optimal transport, arXiv:2504.11141.
- Harris & Tailleux (2018), Assessment of algorithms for computing moist available potential energy, QJRMS 144, 1501–1510.
- Hieronymus & Nycander (2015), Finding the minimum potential energy state by adiabatic parcel rearrangements with a nonlinear equation of state: an exact solution in polynomial time, JPO 45, 1843–1857.
- Hoffman (1963), On simple linear programming problems, in *Convexity*, Proc. Symp. Pure Math. 7, AMS, 317–327.
- Huang (2005), Available potential energy in the world's oceans, J. Mar. Res. 63, 141–158.
- Ilıcak, Adcroft, Griffies & Hallberg (2012), Spurious dianeutral mixing and the role of momentum closure, Ocean Modelling 45–46, 37–58.
- Saenz, Tailleux, Butler, Hughes & Oliver (2015), Estimating Lorenz's reference state in an ocean with a nonlinear equation of state for seawater, JPO 45, 1242–1257.
- Stansifer, O'Gorman & Holt (2017), Accurate computation of moist available potential energy with the Munkres algorithm, QJRMS 143, 288–292.
- Su & Ingersoll (2016), On the minimum potential energy state and the eddy size-constrained APE density, JPO 46, 2663–2674.
- Tailleux (2013), Available potential energy density for a multicomponent Boussinesq fluid with arbitrary nonlinear equation of state, JFM 735, 499–518.
- Tailleux (2018), Local available energetics of multicomponent compressible stratified fluids, JFM 842, R1.
- Tseng & Ferziger (2001), Mixing and available potential energy in stratified flows, Phys. Fluids 13, 1281–1293.
- Winters, Lombard, Riley & D'Asaro (1995), Available potential energy and mixing in density-stratified fluids, JFM 289, 115–128.
- Winters & D'Asaro (1996), Diascalar flux and the rate of fluid mixing, JFM 317, 179–193.
- Wong, Tailleux & Gray (2016), The computation of reference state and APE production by diabatic processes in an idealized tropical cyclone, QJRMS 142, 2646–2657.